

**Imaging Products Division
Bio-Rad Laboratories
Life Science Group**

Welcome to
The Discovery Series™ Quantity One®
1-D Analysis Software

Version 4.4 Update Training Guide

Bio-Rad is pleased to be able to distribute this training update as part of our continuing product support and training effort for our Imaging Products. This guide is designed to give you a comprehensive tour of the newest release of Quantity One version 4.4. The guide is not intended to offer full training on the use of the software; please refer to the User Guide, online Help function, and CD Tutorial for detailed descriptions.

With the release of version 4.4, Quantity One software functionality and user interface have been enhanced. With a developmental focus on improved instrument integration, process automation, and data security, it will appeal to high-throughput users in an industrial environment. The new release of Quantity One is compatible with Macintosh OS 9 and Windows 98 SE, NT 4.0, 2000, and XP.

Highlights of new features in Quantity One version 4.4:

- *User Login*, enables tracking of file modifications by user
- *Automation Manager*, a new tool to automate repetitive analysis
- *3D Viewer*, a three-dimensional visualization tool

Many of the new features and enhancements in this version of Quantity One are the direct result of conversations with our customers. Please let us know what you would like to see in the next version of Quantity One by faxing, calling, or e-mailing our Technical Support staff. You can also use the Solobug program (automatically installed with Quantity One) to submit software feature requests and suggestions.

The newest release of Quantity One is available for download at:
www.bio-rad.com/softwaredownloads.

Version 4.4 Update Training Guide

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Quantity One Basic

With the introduction of a new Quantity One Basic mode, users can now take advantage of a number of useful analysis features without the need to purchase a license or enter a password. Functionality such as acquisition instrument control, image editing, and volume quantitation are available free of charge. Additionally, the software can be loaded on an unlimited numbers of computers in any lab environment, allowing access to the basic functionality.

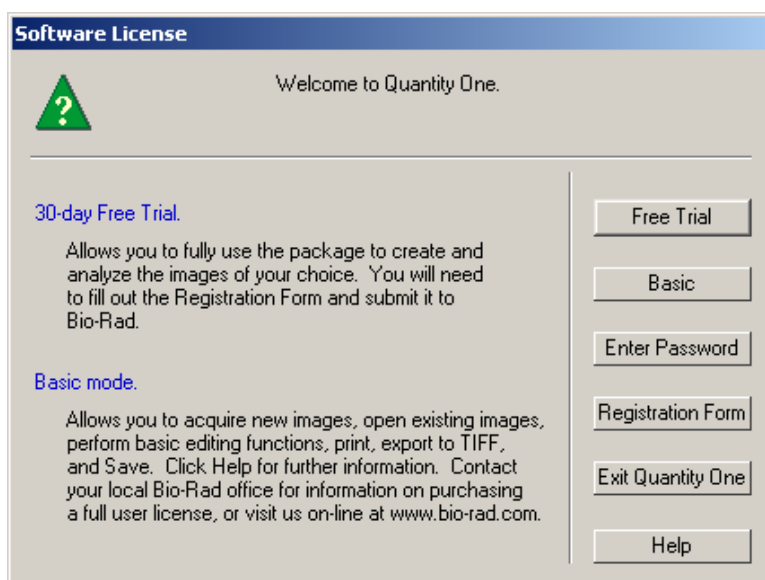


Figure 1. Software License Dialog

When the software is installed (from CD-ROM, catalog # 170-8605) or downloaded (from www.bio-rad.com/softwaredownloads), users are given the following options if a full license password has not yet been entered:

- Click on **Free Trial**, fill out the registration form, and receive a 30-day evaluation license. This allows you to fully evaluate the software and use all sophisticated analysis tools for a limited time period.
- Click on **Basic**. This opens Quantity One in Basic mode. You do not need a software license (password) to run Quantity One Basic. There is no time limit, and Quantity One Basic can be installed on an unlimited number of computers in the lab.

In Quantity One Basic mode, a limited number of the basic functions are activated and the more sophisticated analysis functions are inactivated. In Basic mode, you are able to use the following functions:

- Acquire images from Bio-Rad imaging instruments: the Gel Doc™ 1000/2000, ChemiDoc™, ChemiDoc™ XRS, GS-700™, GS-710™, GS-800™, Fluor-S™, VersaDoc™, Molecular Imager FX™, and Personal Molecular Imager™ imaging systems.
- Open and save images already analyzed with The Discovery Series software.
- Export to TIFF. Open and save TIFF images
- View and edit images using Colors, Zoom, Plot Density, 3D Viewer, Transform, Crop, Flip, and Rotate.
- Add and edit Text Annotations using Text Tool and Line Tool.
- Quantitate bands and objects with Volume Rectangle and Volume Circle Tool, and print and export the Volume Reports.

When Quantity One Basic opens, the new Basic Quick Guide is displayed. It gives you a convenient overview of the features available in Basic mode. To utilize Quantity One's more advanced features such as molecular weight calculations and band pattern matching, a purchased software license is required. The advanced features are grayed out in Quantity One Basic and cannot be activated. You may use the HELP function to learn about the advanced features.

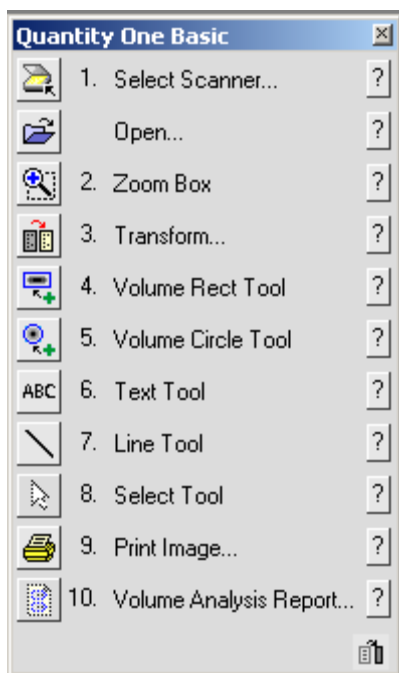


Figure 2. Quantity One Basic Quick Guide

User Login

In order to meet stricter requirements for data security and data integrity, a software user login is implemented in Quantity One. This new feature enables laboratories to record modifications to electronic records, such as the image file. The security level in Quantity One can be set by a software administrator to the following levels:

- No login required. Full file modifications, such as background subtraction, filtering, and cropping, are recorded with date and time stamp.
- Login is required. Full file modifications, such as background subtraction, filtering, and cropping, are recorded by user name and with date and time stamp.
- GLP/GMP mode: No file modification is allowed.

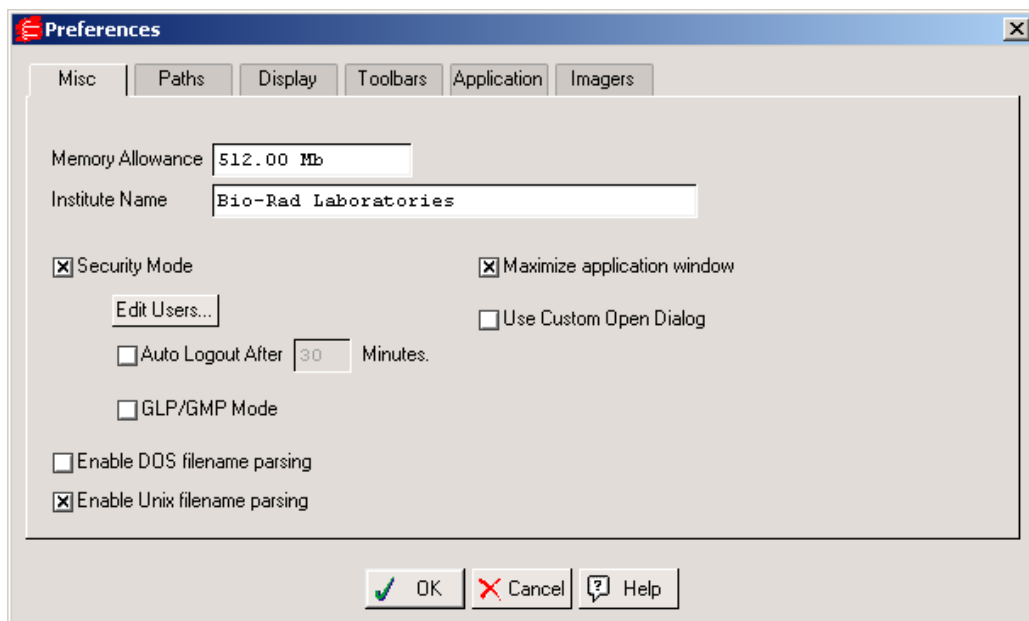


Figure 3. Security Mode is set in Edit>Preferences

When Security Mode is activated, you are prompted to enter a Security Mode Password.

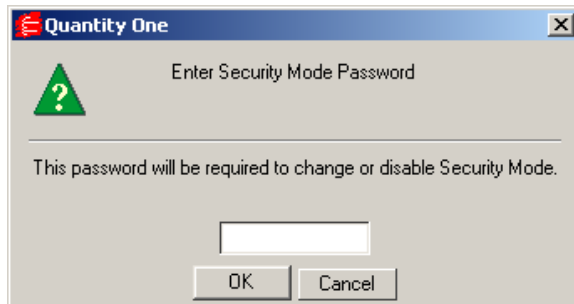


Figure 4. Security Mode Password prompt

The Security Mode Password is required for making changes, adding or removing users, or for disabling Security Mode. It is recommended that a System Administrator be in charge of the Security Mode Password. You must add at least one new user with User Name and Password, or Security Mode will be disabled. Click Edit Users to do this. The Security Mode Password, User Name, and Password are case sensitive.



Figure 5. Add new User

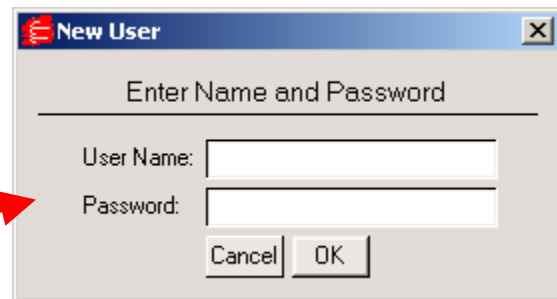


Figure 6. Enter Name and Password

When Security Mode is enabled, all raw data changes and file modifications will be recorded by user and a date and time stamp. Click File>Image Info (or right-click on the image and select Image Info) to display the Image Info.

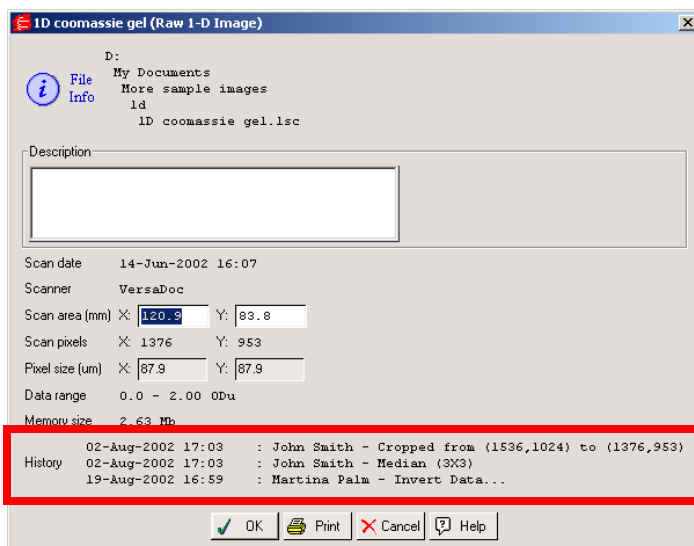


Figure 7. Image Info displays file modifications made by user with a date and time stamp

Security Mode is computer specific, so any images residing on a network or shared drive can be accessed from another computer that does not have Security Mode activated.

If you select Auto Logout, you will automatically be logged out after the specified number of minutes.

After Security Mode is enabled, the GLP/GMP Mode can be enabled. The GLP/GMP Mode allows you to prevent changes to an image that would change the raw image data, such as cropping and filtering the image, subtracting background, or inverting data. To disable GLP/GMP Mode, the Security Mode Password must be entered again.

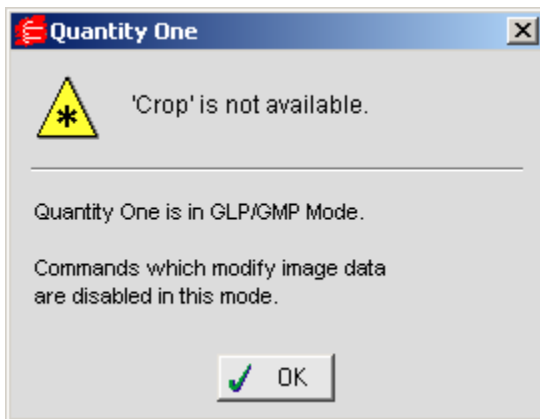


Figure 8. Raw data changes are not permitted in GLP/GMP Mode

Automation Manager

A new tool for automation of repeated analysis has been developed. The powerful Automation Manager allows you to create and save templates for commonly used functionalities:

- Lane and band detection parameters
- Molecular weight determination
- Volume overlays for object quantitation
- Text and lines overlays

The templates are recallable and may be applied on samples of similar types in order to facilitate analysis with little user intervention. Recalled templates can be applied to individual gels or to multiple gels simultaneously.

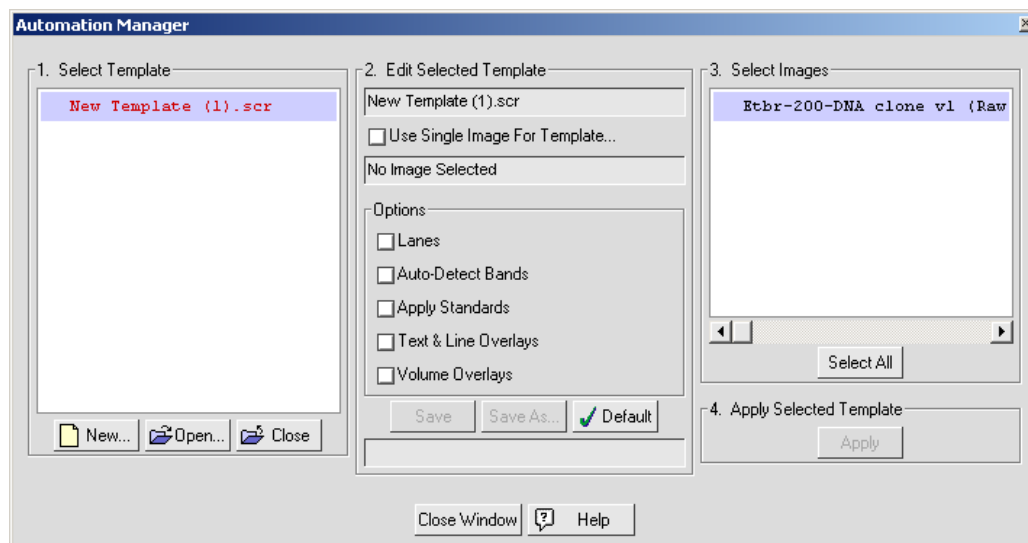


Figure 9. Analysis>Automation Manager

In order for the Automation Manager to work properly for lane-based analysis, the gels to be analyzed should contain the same number of sample lanes and be of similar size. It is recommended that Advanced Crop be used to crop images to identical size.

- Before starting to use the Automation Manager, you would normally already have made the desired analysis on the first image. You must also open the files that you wish to apply a template to.
- Step 1 allows you to save the desired analysis into a Template, or to select a previously created Template.
- In Step 2, you may edit the Template. Checking the box **Use Single Image For Template** allows you to copy the analysis already made from the desired source

image that you have open. This is usually the case. Uncheck the box if you wish to use multiple images as Templates, e.g., when you want to add lane frames and text annotation from different source images.

- Select the options that you want to save in the Template. You are not allowed to select boxes if the option is not already created on the source image, e.g., if you haven't applied Standards on the source image, you cannot select the Apply Standard box.
- You may select Save the Template if you want to use it again later. If you don't save it, the Template will be lost when Quantity One is closed.
- If you choose **Default**, this particular Template will automatically be applied to your image if you select **Apply Default Automation** from the Analysis Menu, without opening the Automation Manager.
- Step 3 allows you to specify which open images you want to apply the Template to.
- Step 4 executes the Automation Manager.

3D Viewer

A new visualization tool has been added that allows display of selected regions of the gel with a three-dimensional view. You can rotate the 3D image vertically and horizontally, and apply different shading textures for an optimal view. This viewing tool requires substantial internal memory of the computer so it is recommended for use only on a smaller area of the sample. The 3D Viewer is a useful tool to determine if a band is actually two or more separated bands.

- Select 3D Viewer from the View menu. Your cursor turns into a crosshair.
- Click and drag your cursor over the image area you would like to view, creating a box. Do not attempt to view a large area of your image as it reduces performance.
- To reposition the box, position your cursor at the center of the box. The cursor appearance will change to a multidirectional arrow symbol. You can then drag the box to a new position.
- To resize the box, position your cursor on a box corner. The cursor appearance will change to a bidirectional arrow. You can then drag that corner in or out, resizing the box.
- Use the Display Options to choose display styles and scale the peaks to desired height.
- The Reset button will return the 3D view back to its initial perspective.

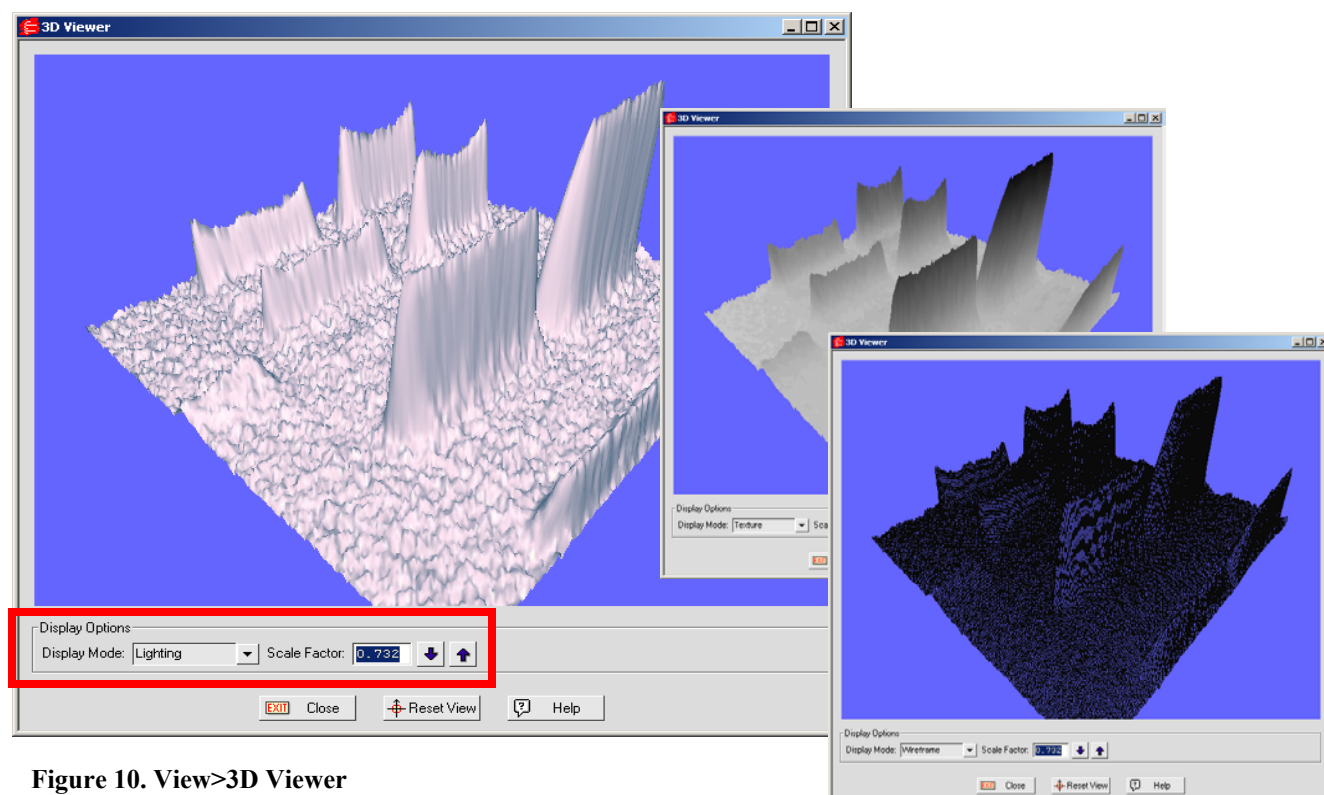


Figure 10. View>3D Viewer

Edit User Label

Volume Labels are convenient to automatically annotate your created Unknown, Standard, and Background Volumes with numbers, without needing to use Text Overlay Tools. In this version of Quantity One, the Volume Label is displayed on the screen and printed out as the default when a volume is created. You may turn the Volume Label off with Volumes>Show/Hide Volume Labels.

The default font size of the text is determined by the actual size of the sample (be aware that imaging systems with zoom lenses are dependent on the user entering correct Image Area under Options). It is now possible to change the font size and style of the Volume Label, using the Edit User Label button in the Volume Properties dialog.

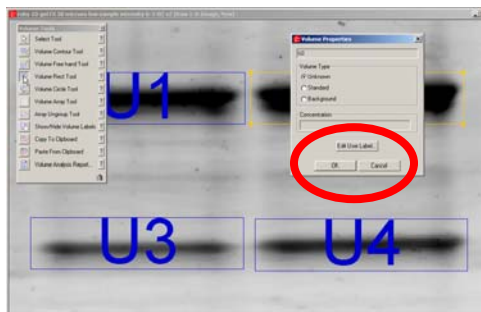


Figure 11. Edit User Label

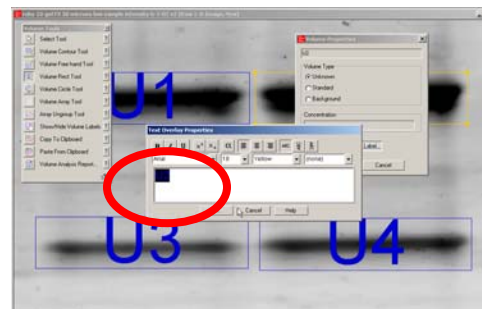


Figure 12. Change the font to desired style

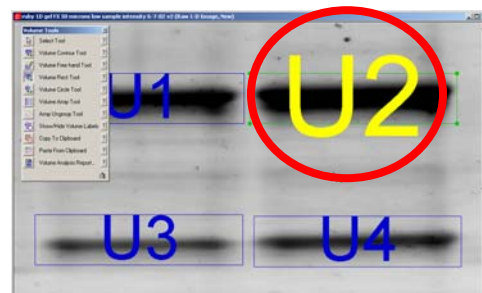


Figure 13. New font style displayed

- Double-click on the Volume Label.
- Click on Edit User Label.
- Change the font to desired style.
- Note that when this feature is used, the Volume Label is converted to a User Label. Consequently, the remaining labels will not automatically be renumbered.

Advanced Crop

To ensure that scans are exactly the same size and shape, use the new Advanced Crop function to save the crop box from one scan and apply it to others. This allows multiplex, multicolor images to be aligned accurately for overlay viewing. It is recommended that you use Advanced Crop when utilizing the Multi-channel Viewer, the Image Stack Tool, and the Automation Manager.

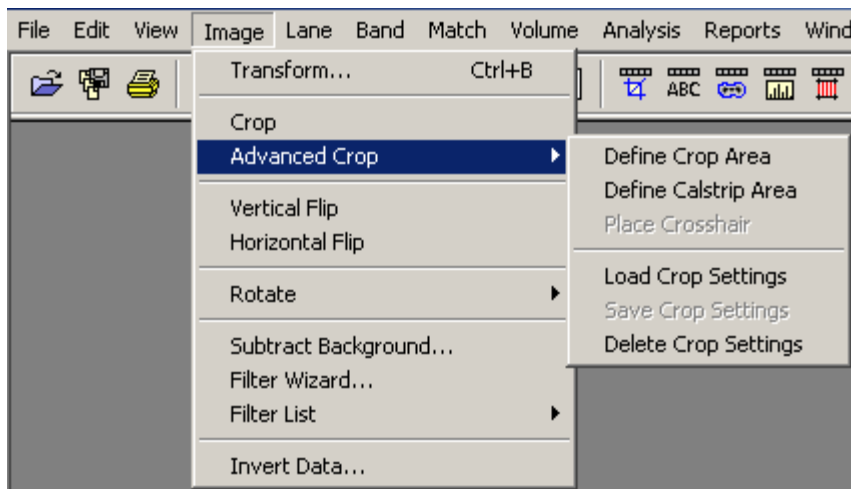


Figure 14. Advanced Crop

- First, define the size of the crop area.
- Then, place the crosshair to facilitate positioning of the crop area in the same place for the next sample. It is recommended the crosshair be aligned with a small image feature that is easily recognizable in all gels to be cropped.
- Save the settings before you execute the crop.
- Open the next image and load the settings. With the placement of the crosshair, you can make sure the position of the crop box is correct.

More Right-Mouse-Click Menu Items

The convenient right-click menu has been expanded with two more options. Click on an open image with your right mouse button (or Ctrl-click on Macintosh) and the menu is displayed.

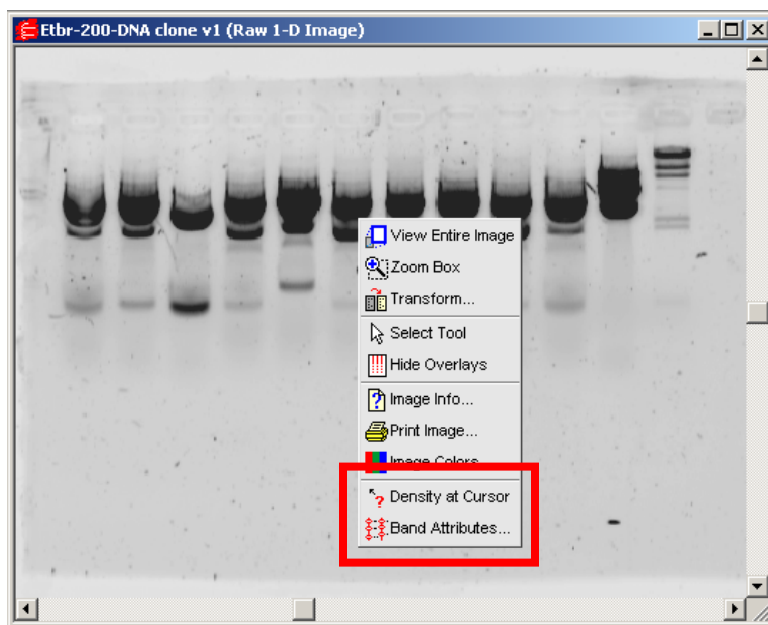


Figure 15. New right-click-menu items

- Density at Cursor allows you to quickly assess the signal intensity of the pixel on the image where you click your mouse.
- Band Attributes allows you to change the information displayed together with your defined bands, such as band number, molecular weight, etc. You may also change how the bands are marked here, as Brackets or Lines.

More Image Colormaps in the Color Palette

New colormaps have been added to display gels with realistic color representation of common stains such as ethidium bromide, SYBR Green, and SYPRO Ruby. In addition, more options to display images in false colors have been added.

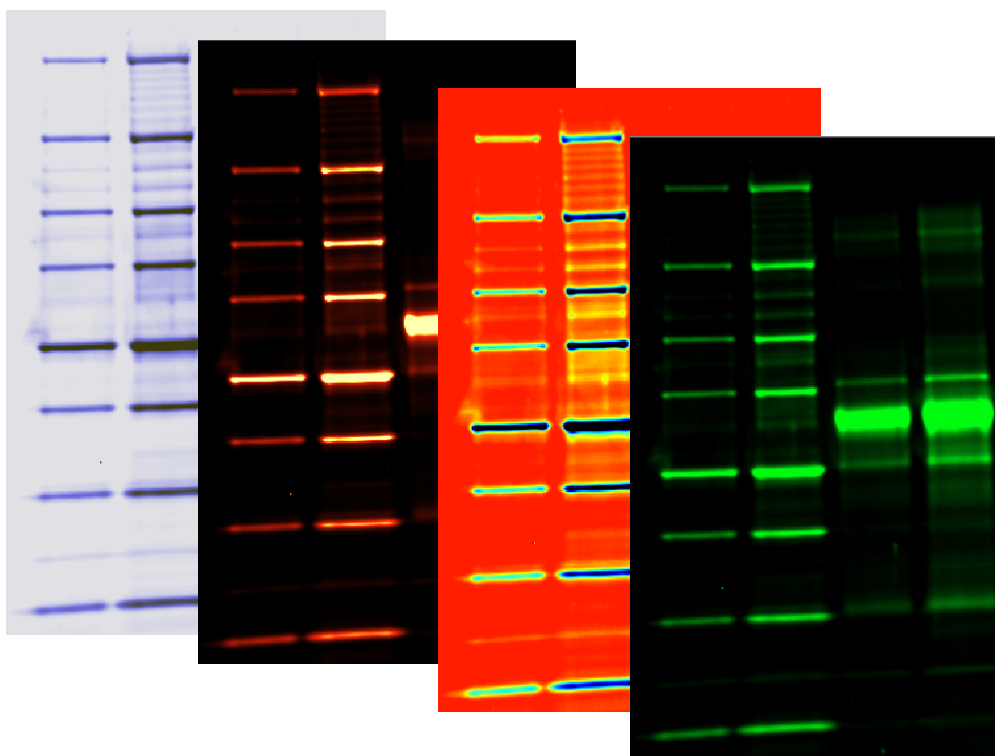
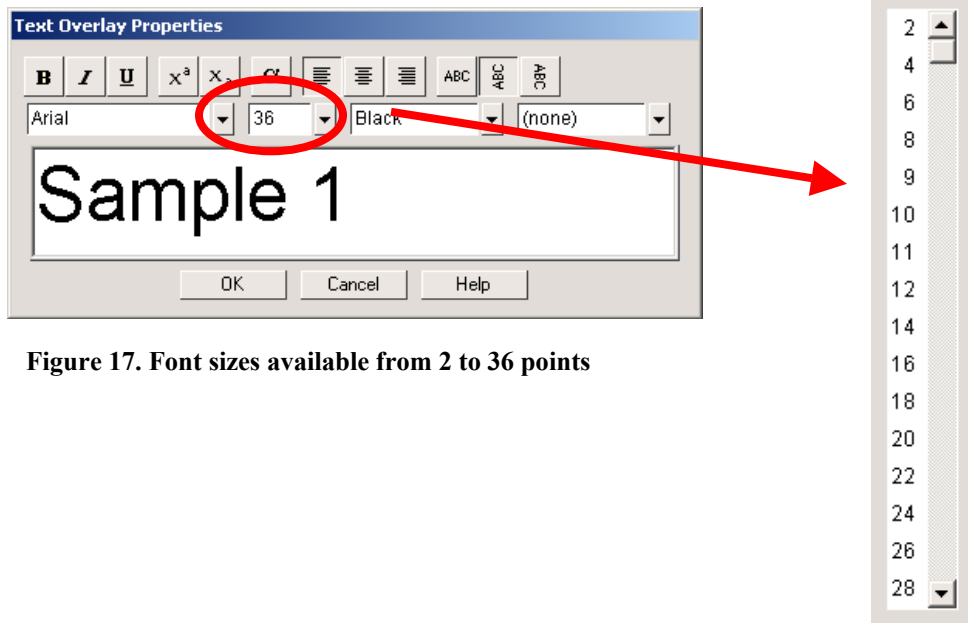


Figure 16. Example of new colors

- Load the new color palette by clicking Edit>Color, or right-mouse-click on an open image and select Image Colors, or change colors from the Transform dialog.
- If the colors are displayed inverted, for example fluorescent dyes from Molecular Imager systems, click Invert Display in the Transform dialog to correct it.

More Font Sizes in the Text Overlays Tools

There is now a wider selection of font sizes available when the Text Overlay Tool is used.



More Archived Molecular Weight Standards

Four new DNA and protein molecular weight standards offered by Bio-Rad Laboratories have been added to the archived list of selectable standards.

- 5 kb ladder
- 8-48 kb standards
- Precision molecular mass ruler
- Precision Plus Protein™ standard

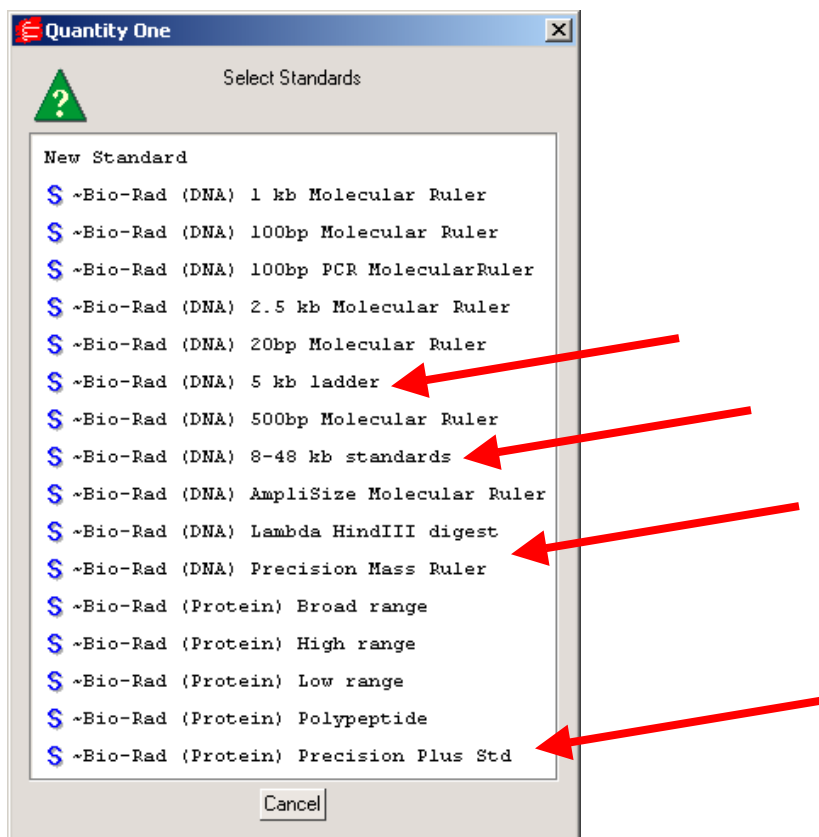


Figure 18. New archived standards

Improved Imaging Instrument Integration: Scanners Included in the Quick Guides

The Quick Guides have been updated to include Select Scanner in Step 1. When using the Quick Guides, you have the choice to scan an image with one of the listed Bio-Rad imagers, or open an already scanned image.

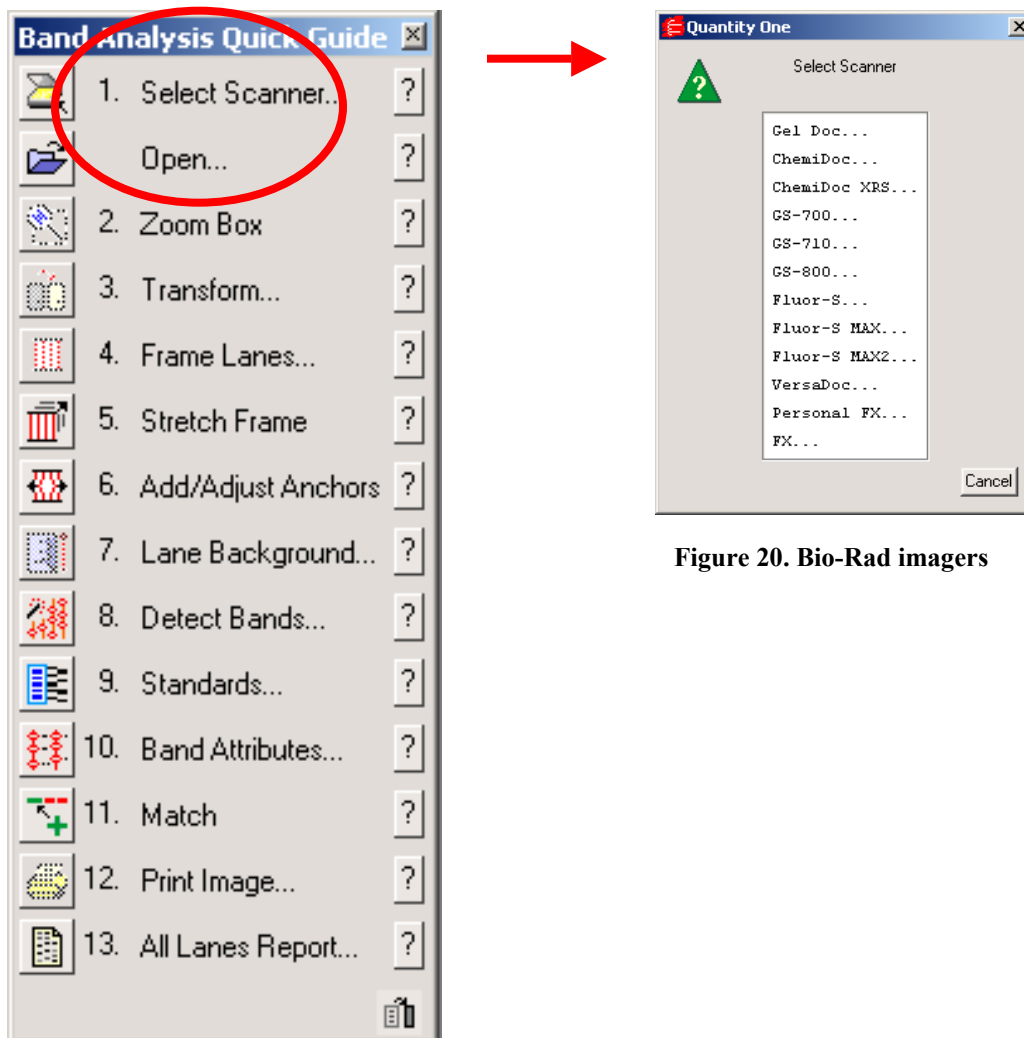


Figure 19. All Quick Guides have Select Scanner as Step 1

Figure 20. Bio-Rad imagers

Improved Gel Doc 2000 and ChemiDoc user interface:

- The motorized lens control buttons now displays text descriptions
- The Image Mode selection step has been moved to Step II
- Rearrangements in Analyze and Select Output buttons to improve workflow

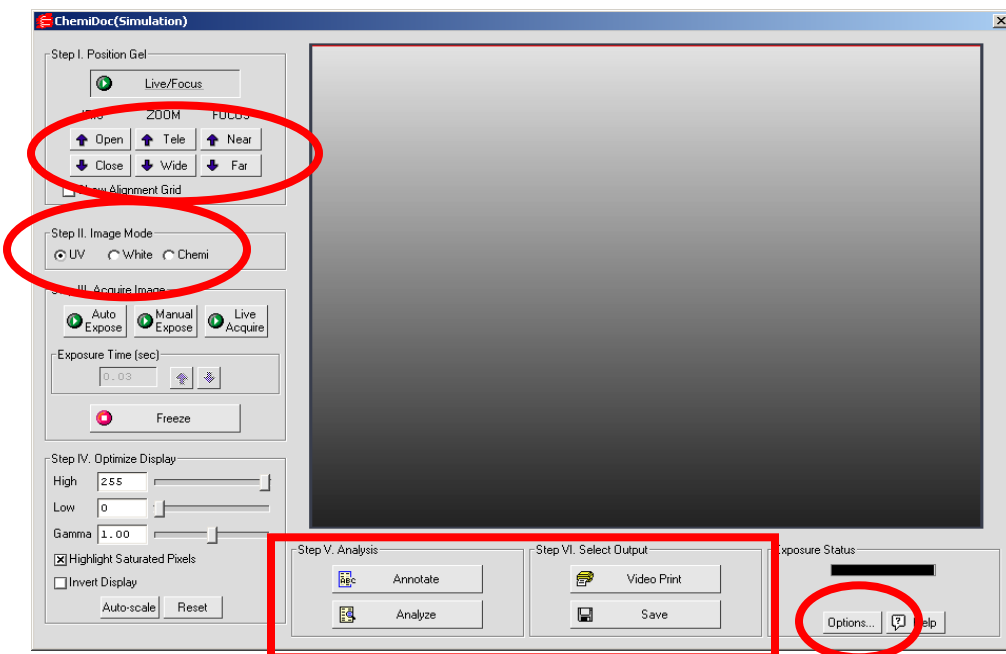


Figure 21. Gel Doc 2000/Chemi Doc improvements

- An optional reminder to the user to remove the filter when using the ChemiDoc in Chemi mode has been added.

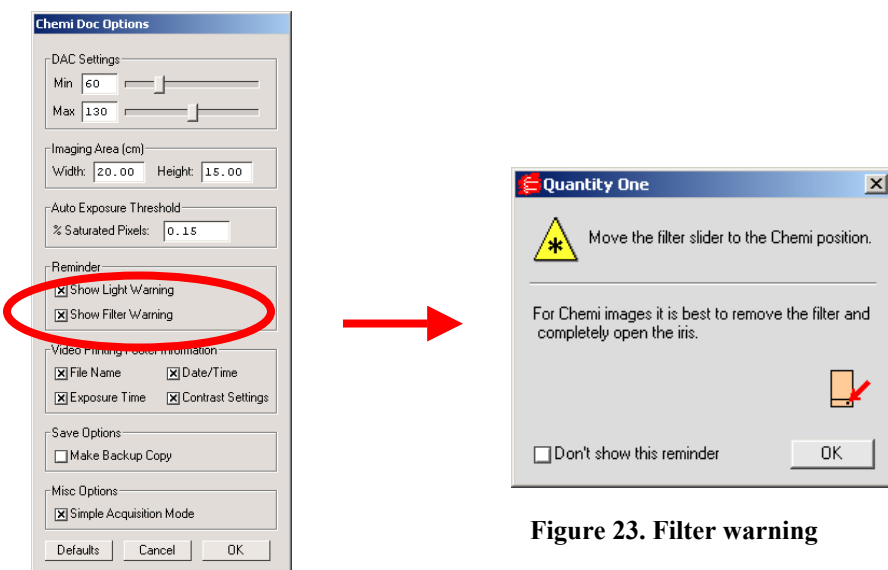


Figure 23. Filter warning

Figure 22. ChemiDoc options

- A Simple Acquisition Mode for the Gel Doc 2000 has been added in order to simplify the workflow for users who are interested in quick Auto Exposure and printing or saving. The Simple Acquisition Mode includes fewer and larger buttons. You may select the Simple Acquisition Mode in Options.

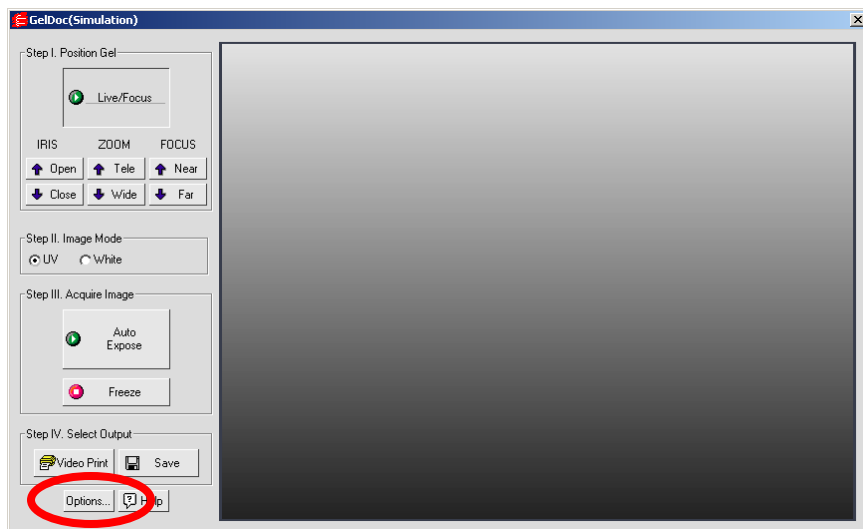


Figure 24. Simple Acquisition Mode for the Gel Doc 2000

- A new imaging instrument, ChemiDoc XRS, is being introduced by Bio-Rad Laboratories. This new instrument offers increased resolution and sensitivity for chemiluminescent images.

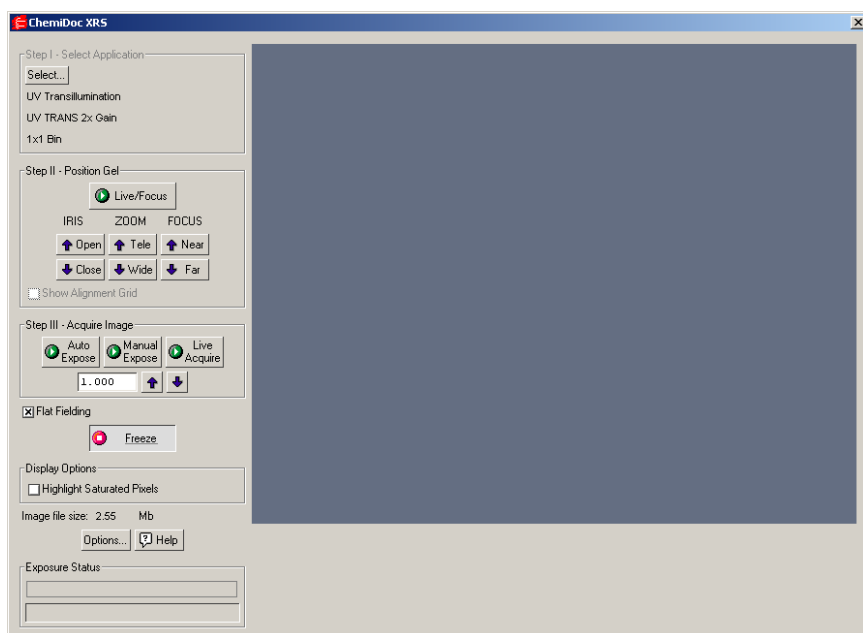


Figure 25. The new ChemiDoc XRS acquisition interface

Improved VersaDoc imaging systems user interface:

- An option to auto-scale the transformation after acquisition has been added. This function is useful for low light applications, to quickly see low signals without having to change the transformation after acquisition.

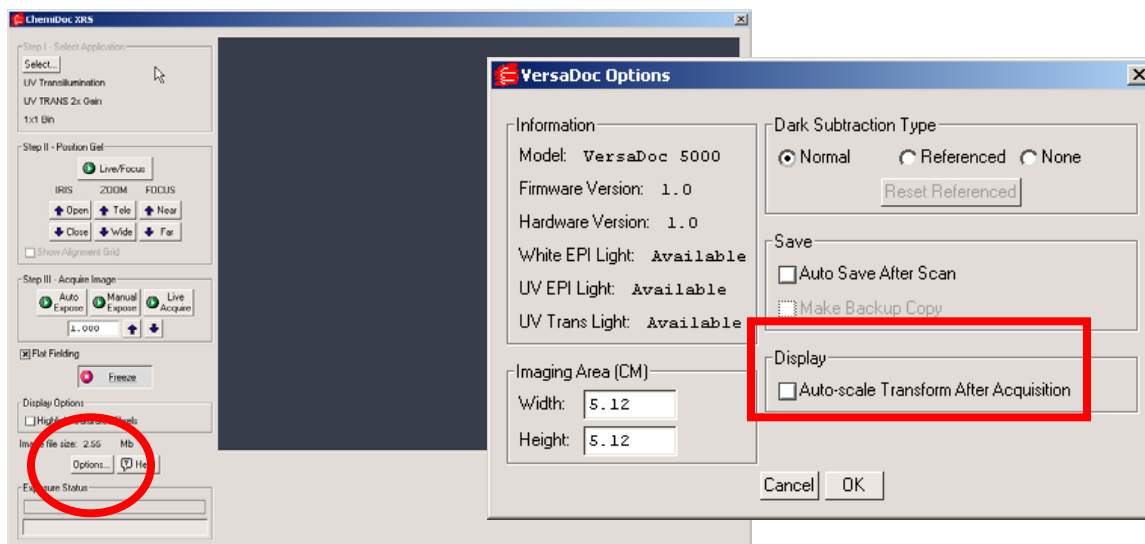


Figure 26. Auto-scale Transformation After Acquisition

Improved Molecular Imager FX systems user interface:

- The currently selected laser and filter settings are displayed in the window in Step 1.
- It is now possible to edit an already created custom application.

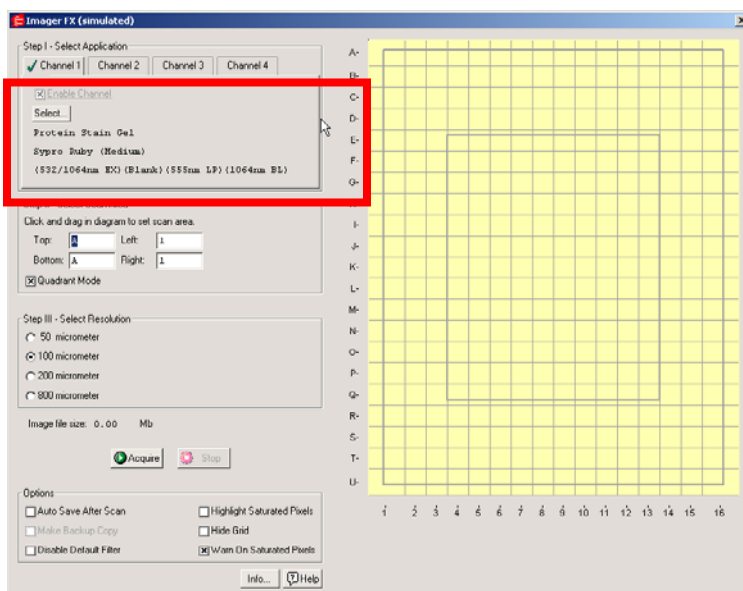


Figure 27. Laser and filter settings displayed

Recommended Computer System Configurations

The Discovery Series Quantity One software version 4.4 is supported on Windows 98 SE, Windows NT 4.0, Windows 2000, and Windows XP operating systems, or on a PowerPC-based Macintosh.

The computer memory requirements are mainly determined by the file size of the images you will scan and analyze. High-resolution image files can be very large. For this reason, we recommend that you archive images on a network file server or high-capacity removable disk.

The following are the recommended computer system configuration for installing and running on a PC:

Operating systems:	Windows 98 SE Windows NT 4.0 with service pack 6 Windows 2000 with service pack 2 or service pack 3 Windows XP
Processor:	Pentium $\geq$ 333 MHz.
RAM:	$\geq$ 128 MB for Gel Doc/ChemiDoc/ChemiDoc XRS, and VersaDoc imaging systems. $\geq$ 256 MB for Molecular Imager FX and Personal Molecular Imager systems, and GS-800 calibrated densitometer.
Hard disk space:	$\geq$ 3 GB.
Monitor:	17" monitor, $\geq$ 1024 x 768 resolution (absolutely required), True Color.
SCSI:	Required for Molecular Imager FX and Personal Molecular Imager systems, and GS-800 calibrated densitometer. Adaptec SCSI card is recommended.

The following are the recommended computers system configuration for installing and running on a Macintosh:

- Operating system:** System 9.0 or higher, excluding Mac OS X.
- Processor:** PowerPC G3 processor or better.
- RAM:** ≥ 256 MB for all Bio-Rad imaging systems.
- Hard disk space:** ≥ 3 GB.
- Monitor:** 17" monitor, $\geq 1024 \times 768$ resolution (absolutely required), Millions of colors.
- SCSI:** Required for Molecular Imager FX and Personal Molecular Imager systems, and GS-800 calibrated densitometer. Adaptec SCSI card is recommended.
- Note:** The default amount of memory assigned to the application on the Macintosh is 128 MB. If the total RAM in your Macintosh is 128 MB or less, you should reduce the amount of memory assigned to the program to 10 MB less than your total RAM. With the application icon selected, go to File > Get Info in your Finder to reduce the memory requirements for the application. See your Macintosh computer documentation for details.

Ordering Information

Catalog number	Description
170-8601	Quantity One Software, PC
170-8609	Quantity One Software, Mac
170-8625	Quantity One Software Upgrade, PC, Mac (for existing users)
170-8605	The Discovery Series CD-ROM
170-8624	Quantity One User Guide

The latest release and previous versions of Quantity One, including the User Guide, may be downloaded from the Internet at no charge. Visit www.bio-rad.com/softwaredownloads. Quantity One version 4.4 may be run in Basic mode without a purchased software license.

Upgrading From an Older Version of Quantity One

Typically, the first minor release after your purchase is free of charge. If you purchased version 4.3, you are entitled to version 4.4 and its releases up to 4.4.9, free of charge.

When ordering 170-8625, Quantity One software upgrade, PC, Mac, you will need to contact Bio-Rad Software Registration in order to receive a new password.

Extended Warranty

Bio-Rad offers an extended warranty program that entitles the user to receive all upgrades for a one-year period, free of charge. Please contact your local Bio-Rad office for a quotation.

Network Licenses

The Discovery Series software is also available as a network installation. In a laboratory with multiple software users, it is convenient to install a “floating” license of x number of copies. Please contact your local Bio-Rad office for a quotation.

Version History

Bio-Rad Laboratories has a long history of scientific imaging and analysis software. The Discovery Series family, including Quantity One, has been used by scientists around the world to analyze and quantitate DNA/RNA and proteins since 1985. The modern Windows version of Quantity One was introduced by Bio-Rad in July 1998. Listed below are the major changes for each Bio-Rad release of Quantity One. In addition to adding drivers for new Bio-Rad Imaging Instruments, enhanced functionality is added in each release.

Version	Major changes	Release	Date
4.0	Acquire images from Bio-Rad instruments Run on a PC as a Windows application (32-bit) License password may be obtained via the Internet 8-bit TIFF export Auto lane finding	4.0.1	July 1998
		4.0.2	January 1999
		4.0.3	April 1999
4.1	Array Tools Colony Counting Batch Mode Free Hand Draw Volume Tool Auto-contour Volume Tools Save Molecular Weight Standards Gaussian Modeling of bands	4.1.0	September 1999
		4.1.1	January 2000
4.2	ChemiDoc Live Acquire FX Multi-channel Sequential Scanning Multi-channel Viewer Tool Image Stack Tool Multiple Regression Models for MW Calculation Local Background Subtraction Recallable Report Templates	4.2.0	July 2000
		4.2.1	January 2001
		4.2.2	April 2001
		4.2.3	July 2001
4.3	Export image in TIFF format including overlays Band Type Presence/Absence Report Convenient right-mouse-click menu Text Tool remembers the last font, color, size, and orientation used	4.3.0	January 2002
		4.3.1	June 2002
4.4	Quantity One Basic mode User Login Automation Manager 3D Viewer Edit User Label Advanced Crop	4.4.0	October 2002

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